www.tntextbooks.in

### 5.13.2 Family: Apocynaceae (milk weed family) (including Asclepiadaceae)

**Systematic position**

| APG classification | | Bentham and Hooker classification | |
|---|---|---|---|
| Kingdom | Plantae | Kingdom | Plantae |
| Clade | Angiosperms | Class | Dicotyledonae |
| Clade | Eudicots | Sub-class | Gamopetalae |
| Clade | Asterides | Series | Bicarpellatae |
| Order | Gentianales | Order | Gentianales |
| Family | Apocynaceae | Family | Apocynaceae |



**Diagnostic features**

- Plants with milky sap.
- Leaves opposite or whorled, exstipulate.
- Flowers pentamerous.
- Stamens epipetalous, connate with corona.
- Fruit a follicle.
- Stigma is thick and massive often connate to stamen to form Gynostegium
- Seeds often with coma (hair).
- Presence of nectariferous disc.

**General Characters**

**Distribution:**

This family is represented by 345 genera, 4,675 species. Mostly tropical and subtropical whereas a few species found in temperate region.

**Habit:** Tree (*Alstonia*), shrub, (*Nerium)*, herb (*Catharanthus)*, woody twiner (*Allamanda*, Succulent, *Adenium* usually twining shrubs with milky sap in laticiferous vessels.

**Root:** Branched tap root system

**Stem:** The stems are succulent in some taxa (*Stapelia, caralluma*), usually erect, branched solid, glabrous, rarely tube like and thick.

**Leaves**: Simple, undivided, sometimes reduced, exstipulate, opposite decusate (*Calotropis*) or sometimes alternate (*Thevetia*) or in whorls of 3 (Nerium), entire, rarely stipulate (*Tabernaemontana*).

**Inflorescence**: A Panicle, dichasial cyme, often umbelliform in (Asclepiadoids) or raceme, or axillary cluster of two flowers each (*Catharanthus*).

**Flowers:** Bracteate, bracteolate, pedicellate, complete, bisexual, actinomorphic, zygomorphic in *Ceropegia* heterochlamydeous , pentamerous, hypogynous but rarely perigynous or epigynous *(Plumeria)*



*Ceropegia spp*

**Calyx:** Sepals 5, synsepalous (at least basally) sometimes aposepalous (*Catharanthus*) deeply lobed ; valvate or quincuncial (*Thevetia*), odd sepal posterior, glandular appendages (*Squamellae*) present on the adaxial side.

**Corolla:** Petals 5, sympetalous united